

IPS18 – Automated Healthcare Document Processing and Intelligence Platform

Industry Domain: Healthcare & Life Sciences

Business Background: Healthcare organizations manage large volumes of patient records, lab reports, prescriptions, and insurance documents. Manual processing is time-consuming, error-prone, and makes retrieving critical patient information difficult. An AI-powered platform can automate document processing, improve information accessibility, and enhance operational efficiency.

Problem Statement: Design and develop an AI-powered Healthcare Document Intelligence Platform that automatically extracts, classifies, summarizes, and enables intelligent search over healthcare documents using AI and NLP techniques.

Current Challenges:

- Manual processing of healthcare documents.
- Difficulty retrieving relevant patient information.
- High risk of errors during document handling.
- Slow access to clinical information.

Project Objectives:

- Automate document extraction and classification.
- Generate AI-based summaries of medical documents.
- Enable intelligent document search and Q&A.
- Improve efficiency and reduce manual effort.

Scope of the Project:

- **In Scope:** PDF and scanned document processing, OCR, information extraction, document summarization, AI-based search assistant, and reporting.
- **Out of Scope:** Medical diagnosis, treatment recommendations, replacement of healthcare professionals, and clinical decision-making.

Input Data Required: Patient records, medical reports, lab reports, prescriptions, insurance documents, and scanned healthcare records.

Expected Output:

- Extracted patient, diagnosis, medication, and test information.
- AI-generated document summaries.
- Healthcare document search assistant.
- Intelligent Q&A over medical documents.

Deliverables:

- Healthcare document processing pipeline
- Information extraction & summarization module
- RAG-based document search assistant
- Source code
- System architecture
- Final report and presentation